

# Tennis Weather Coach - A100 Model Book

A compact neural model trained on NVIDIA A100 to map weather and tennis context into coaching decisions.

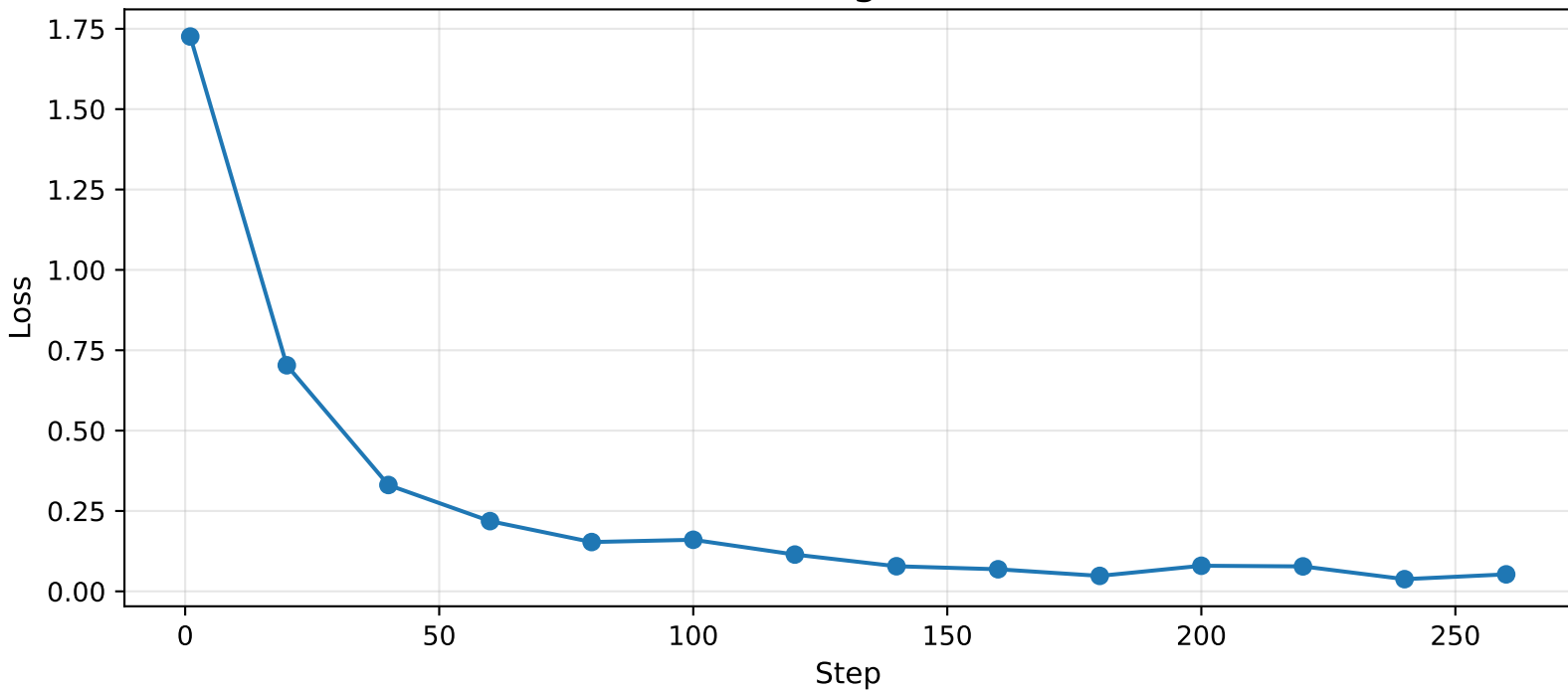
## Executive Summary

- Final action accuracy: 0.978
- Final loss: 0.0529
- Average synthetic weather risk: 0.741
- Hydration estimate: 1.65 liters/hour
- Peak GPU memory during export: 8.37 GB

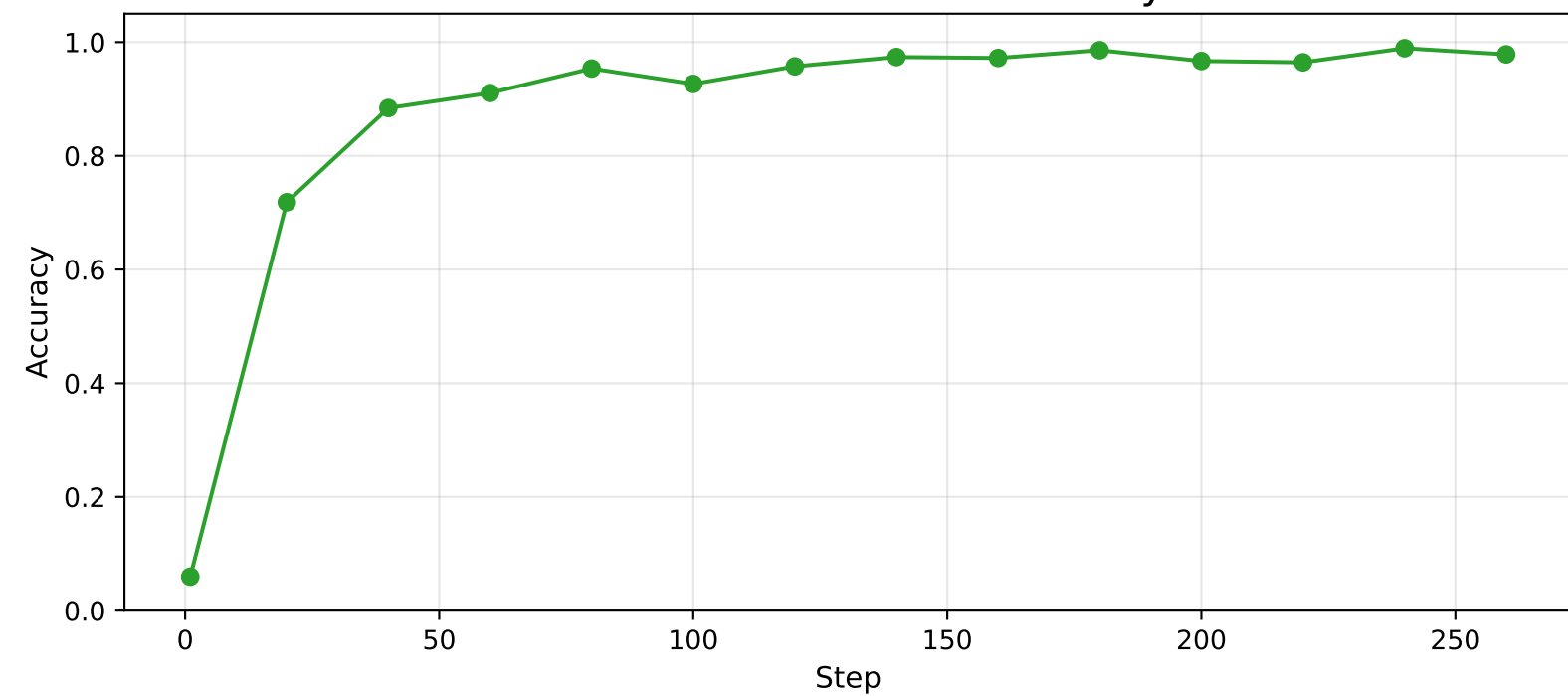
## Why this model matters for tennis

- Tennis coaching and match prep are weather-sensitive: heat, humidity, UV, wind, and rain change risk and tactics.
- The model learns a decision surface for whether to train fully, reduce intensity, move indoor, or postpone.
- The output can become a daily coach assistant once connected to real court-level forecasts.

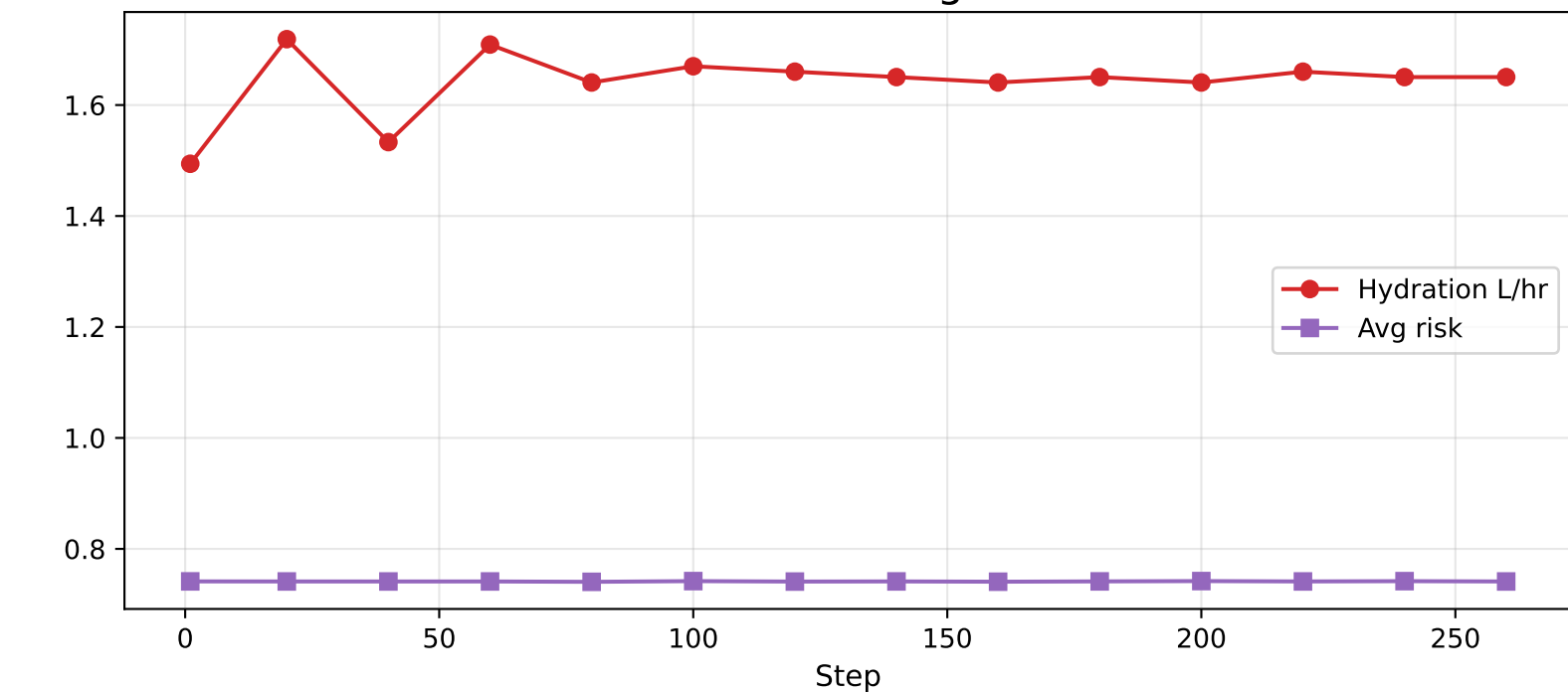
### Training Loss



### Action Classification Accuracy



### Coach-Side Signals



# Model Card

## Inputs

- temp\_c, humidity, wind\_kph, rain\_mm, UV, cloud cover, hour of day
- court heat, player fatigue, match pressure, junior group, indoor option
- derived heat index and total risk

## Outputs

- action class: full training, reduce intensity, technique/recovery, move indoor, postpone/protect match day
- regression heads: session intensity, hydration estimate, rally/session minutes

## Architecture

- TennisWeatherCoach compact MLP
- width: 384
- residual blocks: 5
- checkpoint: tennis\_weather\_coach\_compact.pt

## Training Evidence

- started\_at: 2026-06-14T14:55:34Z
- finished\_at: 2026-06-14T14:55:52Z
- final step: 260

# Coaching Interpretation

## Practical use cases

- Before lesson: choose full session vs technique/recovery based on heat, rain, and wind.
- Before match: protect match-day readiness when heat risk and fatigue are high.
- Junior coaching: raise caution earlier for heat and hydration.
- Outdoor court planning: decide whether an indoor option should be used.

## Example policy meaning

- High heat + high humidity + high UV -> reduce intensity or protect match day.
- Rain risk + indoor option -> move indoor instead of cancelling.
- Strong wind -> reduce rally length and emphasize footwork/serve target drills.
- Low risk + low fatigue -> full training is acceptable.

# Limitations and Next Steps

## Limitations

- This export uses synthetic labels from a transparent coaching rule, not court-specific historical outcomes yet.
- It is a first model artifact, not medical advice. Heat illness decisions still need human judgment.
- Accuracy is measured against the synthetic rule target; real-world validation is still required.

## Next steps

- Connect real forecast data for the actual tennis court location.
- Add player profile: age, fitness, injury, hydration history, and match importance.
- Collect lesson outcomes: completed session, fatigue score, heat symptoms, match result.
- Convert output into a daily coach card: train / adjust / indoor / postpone.